Architecture and Rapid Design Space Exploration of Chiplet-Based NAND Storage Subsystems

A thesis submitted in fulfilment of the requirements

for the degree of Master of Technology

by

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Certificate

It is certified that the work contained in this thesis entitled **"Architecture and Rapid Design Space Exploration of Chiplet-Based NAND Storage Subsystems"** by **Antony Amburose** has been carried out under my supervision and that it has not been submitted elsewhere for a degree.

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Declaration

This is to certify that the thesis titled **"Architecture and Rapid Design Space Exploration of Chiplet-Based NAND Storage Subsystems"** has been authored by me. It presents the research conducted by me under the supervision of **Dr. Kankat Ghosh** .

To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree. Further, due credit has been attributed to the relevant state-of-the-art and collaborations with appropriate citations and acknowledgments, in line with established norms and practices.

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Abstract

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The increasing demand for high-performance and scalable storage in modern computing systems has motivated the exploration of chiplet-based architectures beyond traditional compute and memory subsystems. While chipletization has been widely adopted in processors and high-bandwidth memory systems, its application to NAND-based storage architectures remains relatively unexplored. Conventional UFS and NVMe storage subsystems integrate the controller and NAND arrays within tightly coupled monolithic designs. Transitioning to a chiplet-based storage subsystem—where the controller and NAND arrays are implemented as separate dies interconnected through a high-speed die-to-die interface[1]introduces new architectural trade-offs involving latency, bandwidth, contention, and scalability.

This thesis investigates the architectural implications of chiplet-based partitioning in UFS/NVMe[6, 5] storage subsystems. Inspired by rapid inter-chiplet analytical modeling methodologies

The work develops a scalable modeling framework that represents the controller and NAND chiplets as nodes in a parameterized interconnect graph. Analytical latency and throughput proxy models are derived to capture serialization delay, arbitration effects, and link

saturation behavior without relying on cycle-accurate simulation.

The proposed framework [2] enables systematic design space exploration across key architectural parameters, including number of NAND chiplets, die-to-die link bandwidth and latency, internal parallelism, and request size distributions. Performance metrics such as average latency, tail latency, aggregate throughput, link utilization, and scalability efficiency are evaluated to identify bottlenecks and optimal operating regions.

The outcomes of this research provide quantitative insight into the scalability limits of chiplet-based storage subsystems and establish architectural guidelines for future heterogeneous storage integration. The methodology enables early-stage architectural evaluation with significantly reduced computational complexity, making it suitable for rapid exploration of emerging chiplet-based storage designs.

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Dedicated to family.

Chapter 1

Introduction

1.1 Background

Over the last few decades, semiconductor scaling has enabled integration of increasingly complex systems onto a single die. However, as technology nodes shrink further, the benefits of monolithic System-on-Chip (SoC) design are gradually diminishing. Larger dies suffer from lower yield, higher fabrication cost, reticle size limitations, and increased design complexity. As a result, chiplet-based architectures have emerged as a practical and scalable alternative.

In a chiplet-based design, a large system is partitioned into smaller functional dies—called chiplets—that are interconnected using high-speed die-to-die interfaces such as UCIe [1]. This modular approach offers multiple advantages:

- Improved manufacturing yield due to reduced die area
- Flexibility in mixing process nodes
- Reuse of validated IP across product families
- Scalable architecture through replication

Chipletization has already been adopted in modern processor architectures (e.g., AMD chiplet-based CPUs [3]) and high-bandwidth memory systems. However, NAND-based storage subsystems—such as UFS and NVMe devices [6, 5]—are still largely implemented using tightly integrated monolithic designs.

This thesis investigates what happens when NAND storage is re-architected using chiplet principles.

1.2 Motivation

Traditional NAND storage systems integrate:

- Flash Translation Layer (FTL) controller
- Channel controllers
- NAND interfaces
- within a single silicon die.
- When partitioned into chiplets:
- Controller resides in one die
- NAND arrays reside in separate chiplets
- Communication occurs over a die-to-die link

This architectural transformation introduces new latency components and potential bandwidth bottlenecks.

Key concerns include:

- Serialization delay across die-to-die links
- Link contention under high parallelism
- Saturation behavior
- Tail-latency degradation
- Scalability inefficiency

There is currently no generalized analytical framework for modeling these effects in chiplet-based storage systems.

1.3 Organization of the Thesis

The thesis aims to:

- Develop a graph-based abstraction of chiplet-based storage architecture.
- Derive analytical latency and throughput proxy models.
- Formulate bottleneck and scalability equations.
- Implement a parameterized design space exploration (DSE) framework using [2].
- Identify architectural guidelines for scalable chiplet storage design.

Chapter 2

Literature Review

2.1 Chiplet-Based Architectures

Chiplet architectures decouple system components into smaller dies connected by standardized interconnect protocols like ucie [1]. Performance modeling of inter-chiplet interconnects has been addressed using graph-theoretic abstraction and analytical latency proxies.

These models represent:

- Chiplets as vertices
- Links as edges
- Weighted graph representations

Latency and throughput approximations are derived using weighted path computations and bottleneck analysis.

However, these models have not been extended to NAND storage systems.

2.2 NAND Storage Architecturess

- NAND storage systems consist of:
- Controller with FTL

- Multiple NAND channels
- Planes, blocks, pages
- Latency components include:
- Command issue time
- Data transfer time
- Page program time
- Block erase time

Parallelism is achieved across:

- Channels
- Dies
- Planes

Introducing chiplet partitioning changes the internal communication topology.

Chapter 3

System Architecture and Modeling

3.1 Architectural Abstraction

The proposed chiplet-based storage system is modeled as a graph:

$$G = (V, E) \tag{3.1}$$

where,

$$V = \{C, N_1, N_2, \ldots, N_k\} \tag{3.2}$$

- C represents the controller chiplet.
- N_i represents the ith NAND chiplet.

$$E = \text{Set of die-to-die interconnect links} \tag{3.3}$$

Each link $e \in E$ is characterized by:

- Bandwidth B_e
- Propagation latency L_e

- Arbitration behavior

This abstraction is inspired by rapid inter-chiplet modeling frameworks [2], but adapted to storage transactions.

3.2 Baseline Monolithic Architecture

In conventional monolithic storage systems, the controller and NAND channels reside on the same die. As a result, interconnect delay is negligible and the total request latency can be approximated as:

$$L_{total} = L_{ctrl} + L_{nand} \tag{3.4}$$

where:

- L_{ctrl} is the controller processing latency,
- L_{nand} is the NAND service time.

Since communication occurs on-die, serialization and propagation overheads are minimal and typically ignored in architectural-level modeling.

sectionChiplet-Based Architecture

In a chiplet-based storage architecture, the controller and NAND arrays reside on separate dies and communicate through a die-to-die interconnect. This introduces additional latency components into the critical path.

The total request latency is expressed as:

$$L_{total} = L_{ctrl} + L_{link}^{fwd} + L_{nand} + L_{link}^{ret} \tag{3.5}$$

where:

$$L_{link}^{fwd} = L_{serial} + L_{prop} + L_{arb} \tag{3.6}$$

Similarly,

$$L_{link}^{ret} = L_{serial} + L_{prop} + L_{arb} \tag{3.7}$$

Here:

- L_{serial} represents serialization delay,
- L_{prop} represents propagation delay across the die-to-die link,
- L_{arb} represents arbitration delay due to contention.

This formulation explicitly introduces link-level effects into the storage critical path and highlights the architectural impact of chiplet partitioning.

Chapter 4

Mathematical Modeling

4.1 Link Serialization Delay

4.2 Link Serialization Delay

For a request of size S bits:

$$L_{serial} = \frac{S}{B_{link}} \tag{4.1}$$

where:

- B_{link} is the link bandwidth (bits/sec).

This term dominates for large transfer sizes.

4.3 Arbitration Delay

4.4 Arbitration Delay

Assuming M concurrent requests competing for the link:

$$L_{arb} \approx (M-1)\frac{S}{B_{link}} \tag{4.2}$$

This approximation assumes uniform scheduling and equal-sized transactions.

sectionNAND Service Time

For NAND operations:

$$L_{nand} = t_R \quad \text{(Page Read)} \tag{4.3}$$

$$L_{nand} = t_{PROG} \quad \text{(Page Program)} \tag{4.4}$$

$$L_{nand} = t_{ERASE} \quad \text{(Block Erase)} \tag{4.5}$$

These parameters are technology-dependent.

4.5 Total Latency Model

The total latency is expressed as:

$$L_{total} = L_{ctrl} + 2\left(\frac{S}{B_{link}} + L_{prop} + L_{arb}\right) + L_{nand} \tag{4.6}$$

This equation highlights that link delay appears twice (forward and return paths).

4.6 Throughput Model

Total throughput is bounded by:

$$T = \min\left(B_{ctrl}, B_{link}, N \cdot B_{nand}\right) \tag{4.7}$$

where:

- N is the number of NAND chiplets

- B_{nand} is the per-chiplet bandwidth

This bottleneck formulation aligns with analytical approaches in chiplet modeling literature [2].

4.7 Scalability Efficiency

Scalability efficiency is defined as:

$$\eta = \frac{T_{measured}}{N \cdot B_{nand}} \tag{4.8}$$

- $\eta = 1$ indicates perfect linear scaling
- $\eta < 1$ indicates link-limited regime

4.8 Queueing Model Integration

Using the M/M/1 queue model [9]:

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} \tag{4.9}$$

Total response time:

$$W_{total} = W_q + \frac{1}{\mu} \tag{4.10}$$

where:

- λ is the arrival rate
- μ is the service rate

Under multi-chiplet configuration:

$$\lambda_i = \frac{\lambda}{N} \tag{4.11}$$

This reduces per-chiplet queue depth but may increase link contention.

Chapter 5

Topology Variants and Scalability Analysis

This chapter evaluates different architectural topology options for chiplet-based NAND storage systems. Each topology presents distinct trade-offs in terms of latency, bandwidth utilization, scalability, and bottleneck behavior.

5.1 Star Topology

In the star topology, a single controller chiplet directly connects to all NAND chiplets through independent die-to-die links. This represents the simplest chiplet partitioning strategy.

5.1.1 Architecture Description

- One central controller chiplet
- N independent NAND chiplets
- Direct controller-to-NAND links

The controller acts as the central coordination point for all storage transactions.

5.1.2 Latency Characteristics

The total latency in star topology is given by:

$$L_{total} = L_{ctrl} + L_{link}^{fwd} + L_{nand} + L_{link}^{ret} \tag{5.1}$$

Since communication is single-hop, no intermediate routing delay is introduced.

5.1.3 Scalability Analysis

Aggregate NAND bandwidth is:

$$B_{agg} = N \cdot B_{nand} \tag{5.2}$$

However, effective throughput is constrained by:

$$T = \min(B_{ctrl}, B_{link}, N \cdot B_{nand}) \tag{5.3}$$

As N increases, the link or controller bandwidth becomes the dominant bottleneck.

5.1.4 Advantages

- Minimal routing complexity
- Lowest possible hop count
- Simple arbitration model

5.1.5 Limitations

- Centralized bottleneck at controller
- Limited scalability for large N
- Increased contention under high concurrency

5.2 Multi-Port Controller Topology

In this topology, the controller includes multiple independent die-to-die ports, each serving a subset of NAND chiplets.

5.2.1 Architecture Description

- One controller chiplet with P independent ports
- NAND chiplets distributed across ports

Total available link bandwidth becomes:

$$B_{total} = P \cdot B_{link} \tag{5.4}$$

5.2.2 Scalability Analysis

Throughput is now bounded by:

$$T = \min(B_{ctrl}, P \cdot B_{link}, N \cdot B_{nand}) \tag{5.5}$$

This topology improves scalability compared to the simple star topology.

5.2.3 Advantages

- Higher aggregate bandwidth
- Reduced contention per port
- Better scaling behavior

5.2.4 Limitations

- Increased controller area and complexity
- Higher PHY overhead

5.3 Hierarchical Topology

In hierarchical topology, an intermediate aggregation chiplet is introduced between the controller and NAND chiplets.

5.3.1 Architecture Description

- Controller chiplet
- Aggregation chiplet
- Multiple NAND chiplets

Communication path:

$$\text{Controller} \rightarrow \text{Aggregator} \rightarrow \text{NAND} \tag{5.6}$$

5.3.2 Latency Model

Total latency becomes:

$$L_{total} = L_{ctrl} + L_{link1} + L_{agg} + L_{link2} + L_{nand} + L_{link2}^{ret} + L_{link1}^{ret} \tag{5.7}$$

Where:

- L_{agg} represents aggregation processing delay
- L_{link1} and L_{link2} represent first and second hop link delays

5.3.3 Scalability Behavior

Hierarchical design improves fan-out scalability. However, additional hop latency may increase average access time.

5.3.4 Advantages

- Improved scalability for large N
- Distributed arbitration

5.3.5 Limitations

- Increased latency due to extra hop
- Additional chiplet overhead

5.4 Comparative Scalability Efficiency

Scalability efficiency is defined as:

$$\eta = \frac{T_{measured}}{N \cdot B_{nand}} \tag{5.8}$$

- $\eta = 1$ indicates ideal linear scaling.
- $\eta < 1$ indicates bottleneck-limited scaling.

Comparative analysis across topologies allows identification of optimal architecture under varying link bandwidth and chiplet count constraints.

Chapter 6

Design Space Exploration Framework

In chiplet-based storage architecture, performance does not scale linearly with the number of NAND chiplets. Beyond a certain point, the interconnect becomes the dominant bottleneck. Therefore, a systematic design space exploration (DSE) methodology is required to understand scalability limits and identify optimal architectural regions.

The exploration approach adopted in this work is inspired by rapid inter-chiplet analytical modeling techniques proposed in [2], where graph-based abstractions and bottleneck analysis are used for scalable architectural evaluation.

6.1 Design Parameters

The following parameters define the architectural design space:

- Number of NAND chiplets (N)
- Link bandwidth (B_{link})
- Link propagation latency (L_{prop})
- Controller bandwidth (B_{ctrl})
- NAND service rate (μ)
- Request size distribution (S)

Each parameter directly influences latency, throughput, and scalability efficiency.

6.1.1 Chiplet Count (N)

Increasing N improves aggregate NAND bandwidth:

$$B_{agg} = N \cdot B_{nand} \tag{6.1}$$

However, beyond a threshold value N^*, link bandwidth becomes limiting:

$$N^* = \frac{B_{link}}{B_{nand}} \tag{6.2}$$

For $N > N^*$, the interconnect saturates and additional chiplets do not contribute to throughput improvement. This bottleneck-driven behavior follows the min-cut principle commonly used in chiplet network modeling [2].

6.1.2 Link Bandwidth (B_{link})

Link bandwidth directly impacts serialization delay:

$$L_{serial} = \frac{S}{B_{link}} \tag{6.3}$$

Higher B_{link} reduces latency and increases the saturation threshold. This aligns with analytical link modeling approaches discussed in [2].

6.1.3 Service Rate (μ)

NAND chiplet service rate determines queue behavior. Request-level contention is approximated using the M/M/1 queue model [**?**]:

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} \tag{6.4}$$

Under multi-chiplet configuration:

$$\lambda_i = \frac{\lambda}{N} \tag{6.5}$$

Increasing N reduces per-chiplet queue depth but may increase interconnect-level contention.

6.2 Exploration Methodology

The exploration framework evaluates configurations through the following structured process:

1. Select parameter set $\{N, B_{link}, L_{prop}, \mu, S\}$
2. Compute total latency using the analytical model
3. Compute throughput using bottleneck formulation
4. Calculate scalability efficiency
5. Store metrics for comparative analysis

This approach avoids cycle-accurate simulation and enables rapid evaluation of thousands of configurations, similar to scalable architectural exploration techniques in [2].

6.3 Performance Metrics

6.3.1 Average Latency

$$L_{avg} = L_{ctrl} + 2\left(\frac{S}{B_{link}} + L_{prop} + L_{arb}\right) + L_{nand} \tag{6.6}$$

6.3.2 Throughput

$$T = \min(B_{ctrl}, B_{link}, N \cdot B_{nand}) \tag{6.7}$$

This bottleneck formulation is consistent with analytical min-bound throughput estimation [2].

6.3.3 Scalability Efficiency

$$\eta = \frac{T_{measured}}{N \cdot B_{nand}} \tag{6.8}$$

6.4 Saturation Boundary Identification

Saturation occurs when:

$$N \cdot B_{nand} \geq B_{link} \tag{6.9}$$

This defines the transition between:

- NAND-limited regime
- Link-limited regime

Identifying this boundary enables architectural optimization and early-stage pruning of non-scalable configurations.

Chapter 7

Scope of Work

7.1 In scope

.Architectural-level modeling of chiplet-based storage subsystems

- Graph-based abstraction of controller–NAND interconnect
- Analytical performance estimation
- Design space exploration
- Comparative analysis of topology variants

7.2 Out of Scope

Out of Scope implementation or layout circuit-level modeling and electromagnetic packaging analysis -specific host modeling

7.3 Expected Analytical Results

1. Chiplet count scaling is not linear due to link contention.
2. Beyond certain N, link becomes dominant bottleneck.
3. Hierarchical topology may outperform star for large N.

4 .Tail latency increases sharply near link saturation.

5 .Optimal chiplet count depends on link BW and NAND timing.

Chapter 8

Conclusions and future work

8.1 Conclusions

This thesis proposes a unified analytical framework for evaluating chiplet-based NAND storage systems ussing rapid chiplet tool[2]. In dissections 2 , the following contribution will be made.

1. Graph-based architectural abstraction
2. Analytical latency derivation
3. Bottleneck-based throughput modeling
4. Scalability efficiency metric
5. Automated design space exploration

8.2 Scope for future work

- Integration with reinforcement learning–based topology search
- Support for UCIe-compliant protocol modeling
- Integration with real workload traces
- Extension to 3D chip stacking
- AI-driven multi-objective optimization

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